

IBM DATA SCIENCE CAPSTONE PROJECT

SpaceX Falcon 9 First Stage Landing Prediction

An end-to-end machine learning pipeline analyzing launch parameters, payload mass, geospatial proximity, and predictive classification algorithms.



COMPLETED NOTEBOOKS & SOURCE CODE

<https://github.com/Jyotibrata-07-Chakraborty/IBM-Data-Science-Course.git>

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• IBM DATA SCIENCE PROFESSIONAL CERTIFICATE



Outline

- **Executive Summary**
- **Introduction**
- **Methodology**
- **Results**
- **Conclusion**

Executive Summary

Data Collection

Data Wrangling

Exploratory Data Analysis

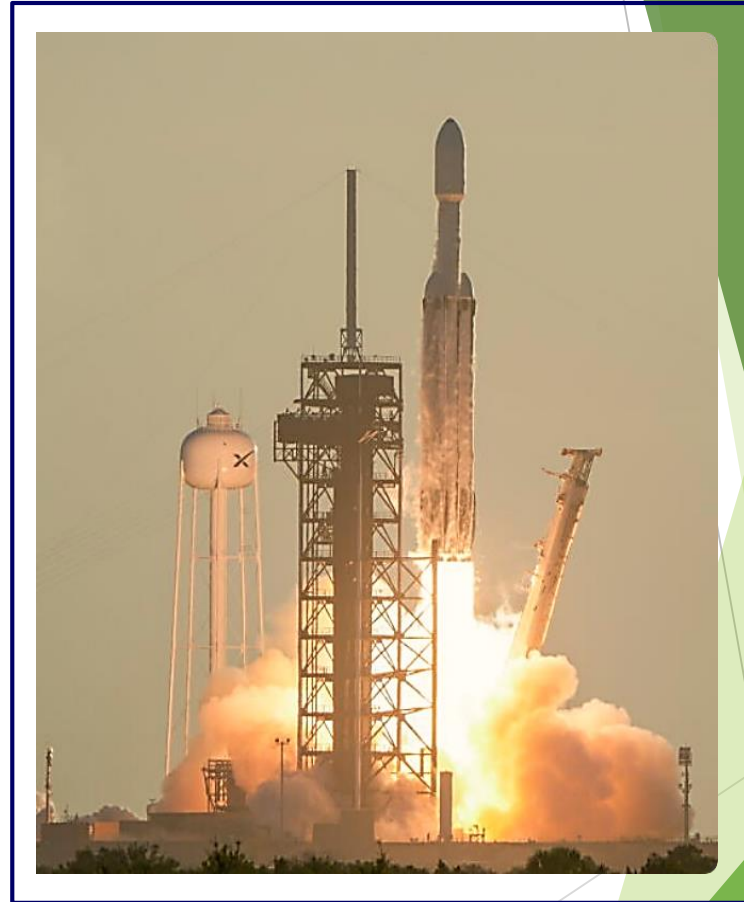
Exploratory Data Visualization

Predictive Analysis- Machine Learning

Introduction

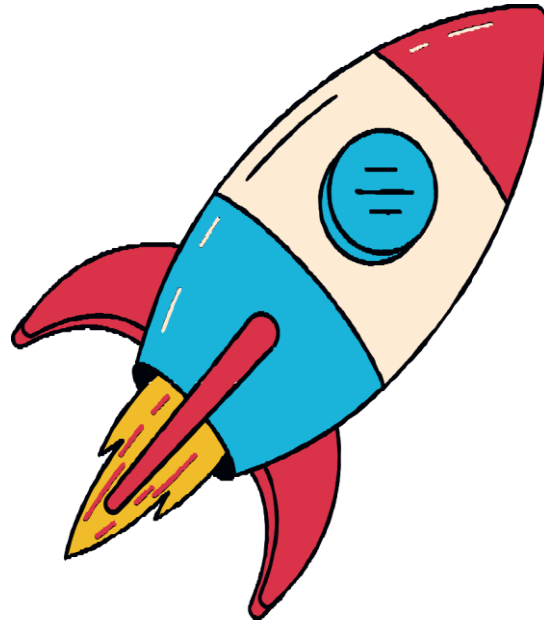
SpaceX's Goal

- Sending spacecraft to the International Space Station (ISS).
- Providing high-speed satellite internet access across the world through Starlink technology.
- Transporting people and cargo into space while advancing space exploration and developing future space technologies.



FALCON 9 ROCKETS

Falcon 9 is a two-stage reusable rocket developed by SpaceX for launching satellites, cargo, and crew into space. Its most important feature is the reusable first stage, which can return to Earth and land either on a landing pad or an autonomous drone ship, helping reduce launch costs. Falcon 9 is widely used for **Starlink satellite deployments, NASA missions, ISS resupply, crewed flights, and commercial satellite launches**, making it a key part of SpaceX's space exploration and transportation efforts.



Data Collection Methodology Flow Chart

Methodology

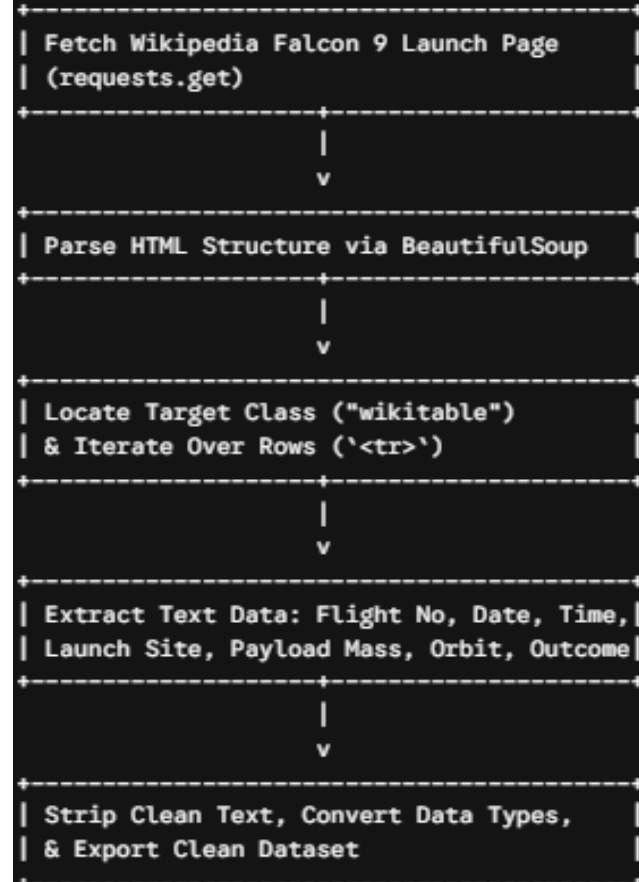
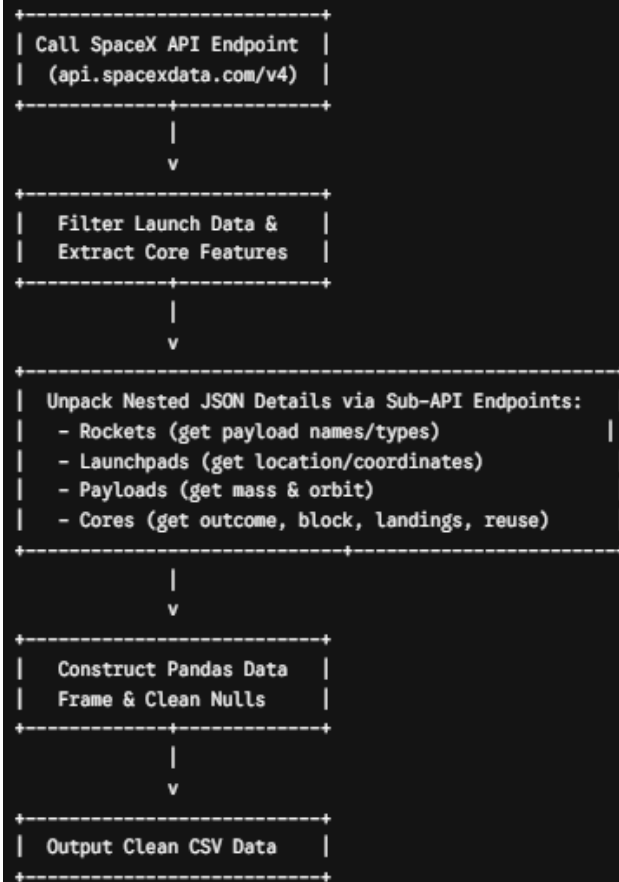
API

spacex_url="https://api.spacexdata.com/v4/launches/past"

https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Falcon_Heavy_launches&oldid=1027686922

Web Page

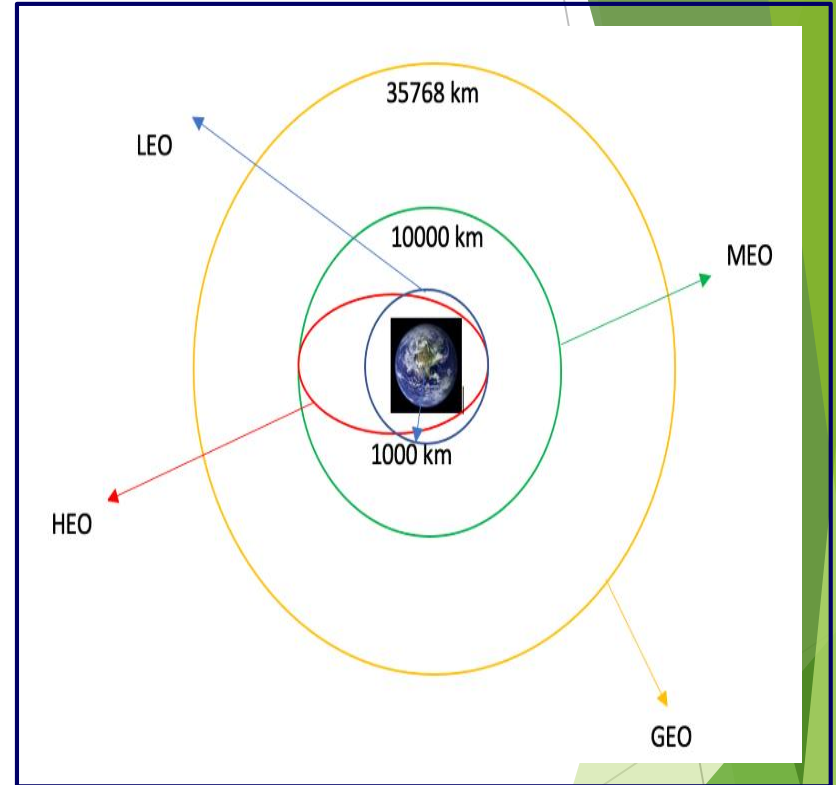
Require the data from SpaceX API



Collect data from a Wikipedia page

Data Wrangling

- Calculate the number of launches at each launch site.
- Calculate the number and frequency of each orbit type.
- Calculate the number and frequency of mission outcomes for each orbit type.
- Create a landing outcome label based on the `Outcome` column.



EDA & Visual Analytics Method



Exploratory Data Analysis

Utilized Matplotlib and Seaborn to analyze bivariate relationships between Payload Mass, Flight Number, Launch Site, and Target Orbit vs. Landing Success.



SQL Relational Querying

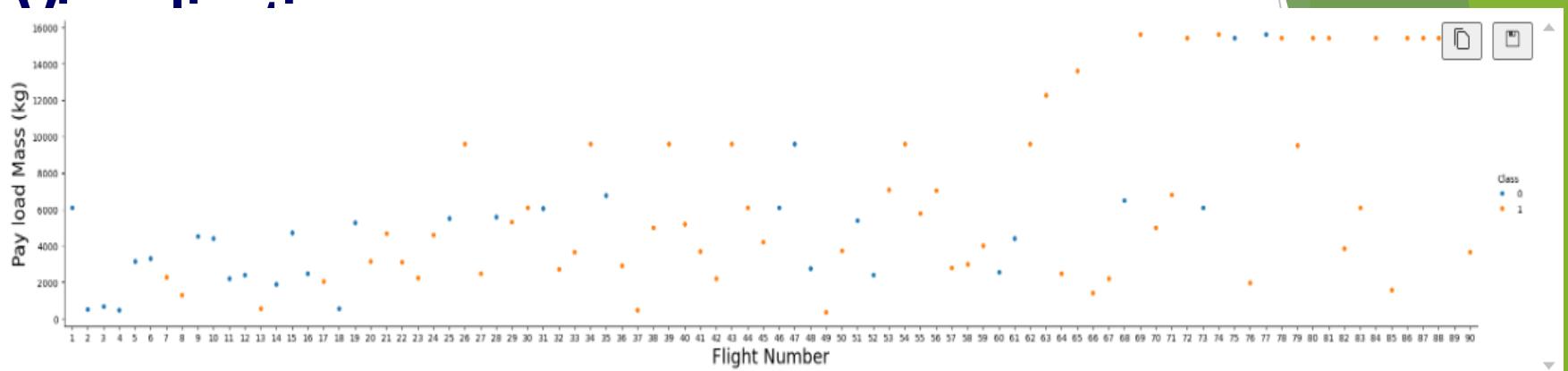
Executed structured SQL queries (SQLite / DB2) to slice payload distributions, rank successful landing boosters, and measure mission counts per site.



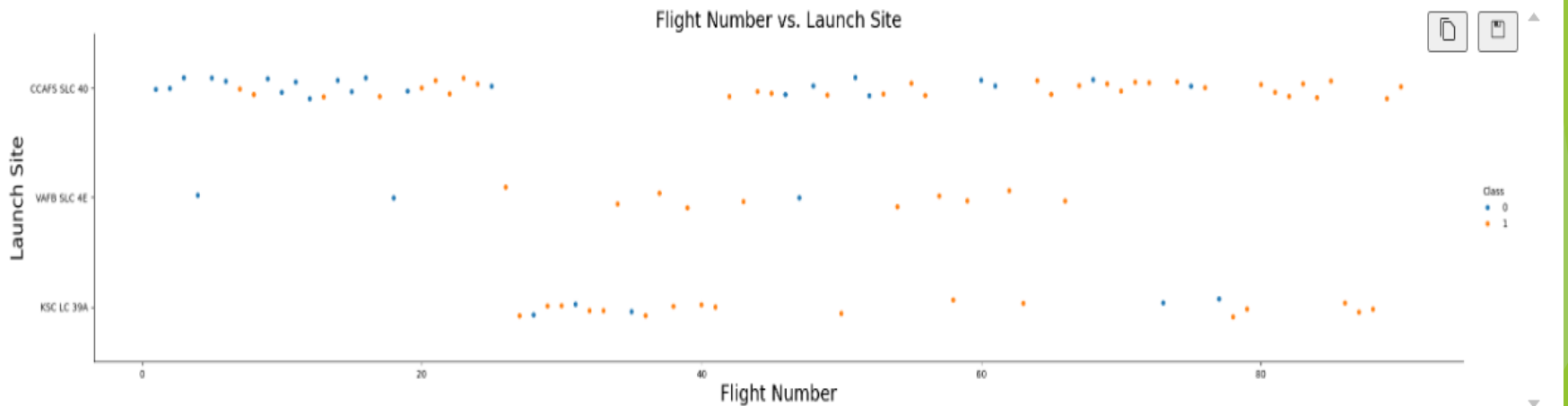
Geospatial & Dashboards

Built interactive Folium maps with distance measurement markers to safety proximities and constructed a real-time Plotly Dash web application with payload sliders.

EDA with Data



It has been identified that different launch sites have different success rates. **CCAFS LC-40** has a success rate of approximately **60%**, while **KSC LC-39A** and **VAFB SLC 4E** have higher success rates of around **77%**.



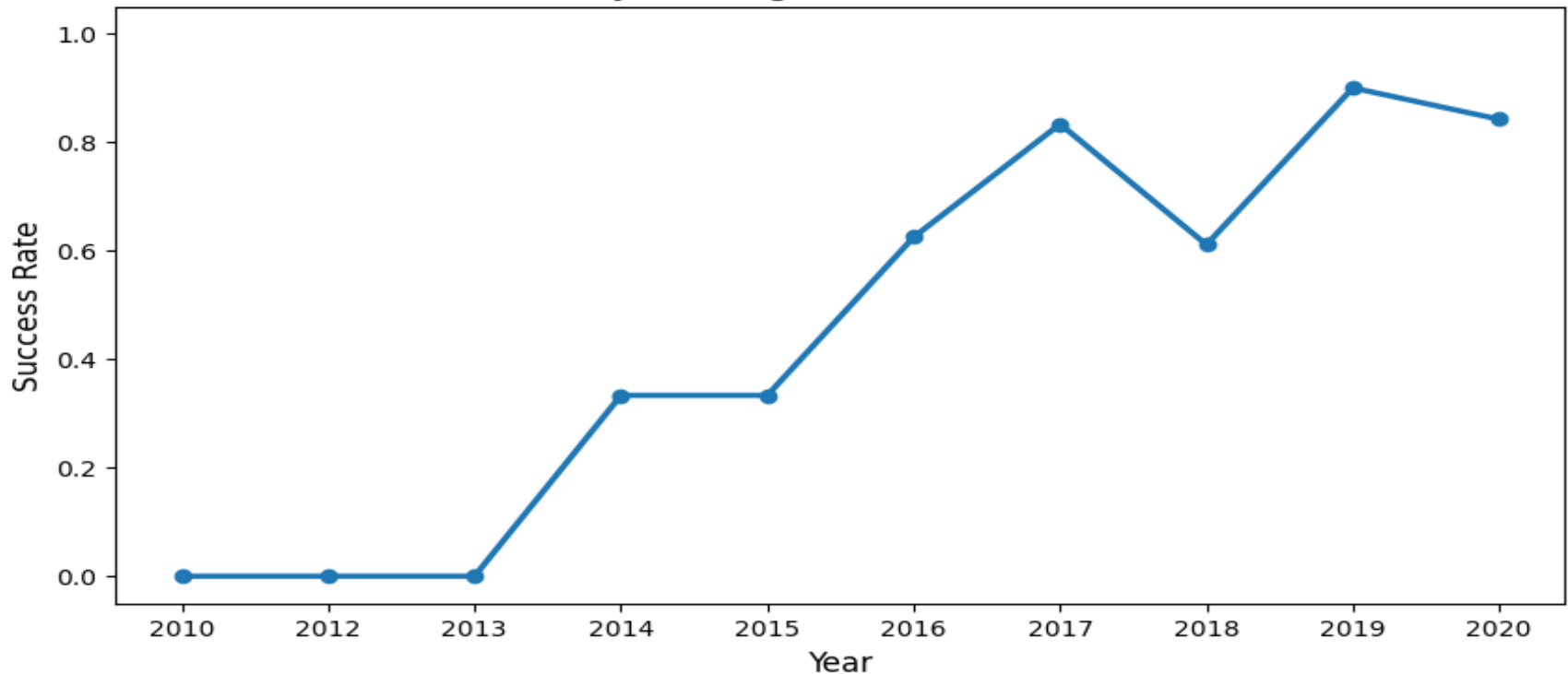
Use the `catplot` function to visualize Flight Number against Launch Site. Set the `x` parameter to `FlightNumber`. Set the `y` parameter to `LaunchSite`. Set the `hue` parameter to `class` to distinguish between successful and

Visualization of the launch success yearly trend

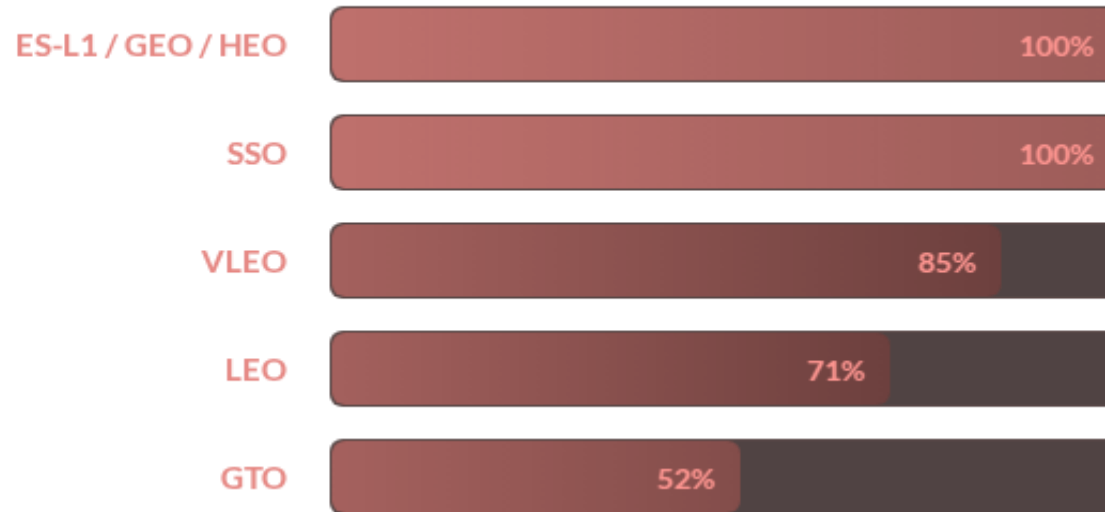
Certainly, Here is your data cleaned up into a readable table:

Index	FlightNumber	Date	BoosterVersion	PayloadMass	Orbit	LaunchSite	Outcome	Flights	GridFins	Reused	Logs	LandingPad	Block	ReusedCount	Serial	Longitude	Latitude	Class
0	1	2010	Falcon 9	6104.96	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	80003	80.577366	28.561857	0
1	2	2012	Falcon 9	525.00	LEO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	80005	80.577366	28.561857	0
2	3	2013	Falcon 9	677.00	ISS	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	80007	80.577366	28.561857	0
3	4	2013	Falcon 9	500.00	PO	VAFB SLC 4E	False Ocean	1	False	False	False	NaN	1.0	0	81003	-120.610829	34.632093	0
4	5	2013	Falcon 9	3170.00	GTO	CCAFS SLC 40	None None	1	False	False	False	NaN	1.0	0	81004	80.577366	28.561857	0

Yearly Landing Success Rate Trend



Success Rate by Orbit Type (%)



Key Scatter & Line Plot Takeaways

- Clear upward trend; early flights (<20) had frequent landing failures, whereas recent flights (>60) achieved over 85% success.
- 🏠 Payloads between 2,000 kg and 5,000 kg bound for GTO show mixed results, while heavy payloads (>6,000 kg) to LEO perform consistently well.

EDA with Data Visualization
- Major Takeaways



EDA SQL Analysis Findings

SQL Analytical Query Focus	SQL Function / Logic	Key Result / Finding
Unique Launch Sites	SELECT DISTINCT Launch_Site	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
NASA CRS Payload Total	SUM(PAYLOAD_MASS__KG_)	45,596 kg delivered for NASA Space Station missions
F9 v1.1 Average Payload	AVG(PAYLOAD_MASS__KG_)	2,928.4 kg average payload mass
First Successful Ground Pad	MIN(Date) WHERE Landing = 'Success'	Achieved on 2015-12-22 (LZ-1 Ground Pad)
Drone Ship Boosters (4k-6k kg)	WHERE Payload BETWEEN 4000 AND 6000	Booster Cores B1022, B1026, B1029, B1031.1

Names of the unique launch sites in the space mission

```
%sql SELECT DISTINCT Launch_Site FROM SPACEXTABLE;
```

```
* sqlite:///my_data1.db
```

Done.

Launch_Site

CCAFS LC-40

VAFB SLC-4E

KSC LC-39A

CCAFS SLC-40

Five records where launch sites begin with the string 'CCA'

```
%sql SELECT * FROM SPACEXTABLE WHERE Launch_Site LIKE 'CCA%' LIMIT 5;
```

```
* sqlite:///my_data1.db
```

Done.

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS_KG_
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677

Total payload mass carried by boosters launched by NASA (CRS)

```
%sql SELECT SUM(PAYLOAD_MASS_KG_) AS Total_Payload_Mass FROM SPACEXTABLE WHERE Customer = 'NASA (CRS)';
```

```
* sqlite:///my_data1.db
```

Done.

Total_Payload_Mass

45596

Average payload mass carried by booster version F9 v1.1

```
%sql SELECT AVG(PAYLOAD_MASS_KG_) AS Average_Payload_Mass FROM SPACEXTABLE WHERE Booster_Version = 'F9 v1.1';
```

```
* sqlite:///my_data1.db
```

Done.

Average_Payload_Mass

2928.4

The date when the first succesful landing outcome in ground pad was acheived.

```
● %sql SELECT MIN(Date) AS First_Successful_Ground_Pad_Landing FROM SPACEXTABLE WHERE "Landing_Outcome" =
```

```
* sqlite:///my_data1.db
```

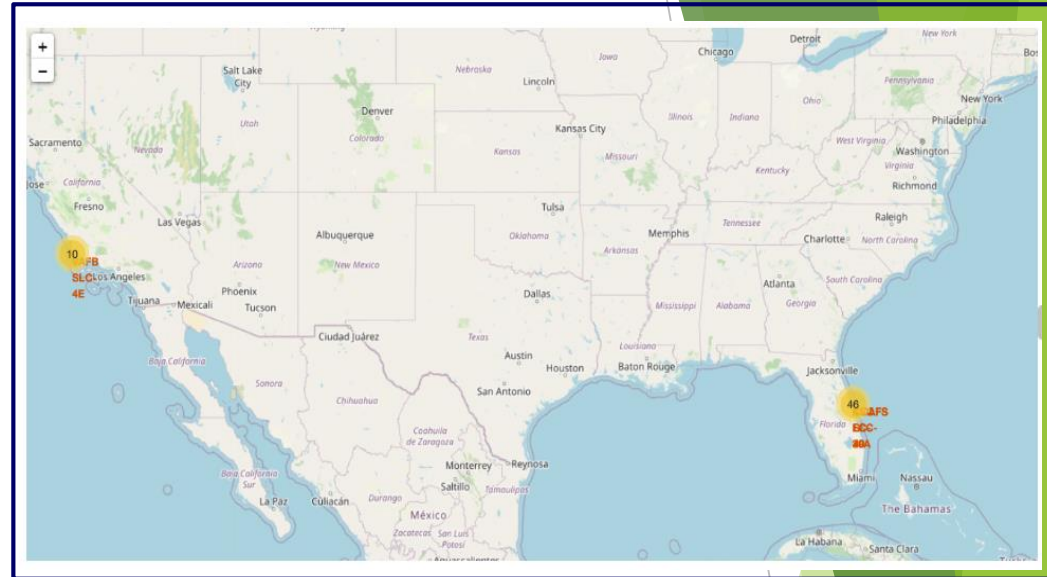
Done.

First_Successful_Ground_Pad_Landing

2015-12-22

Interactive Visual Analytics with Folium

- Mark all launch sites on a map
- Mark the success/failed launches for each site on the map
- Calculate the distances between a launch site to its proximities



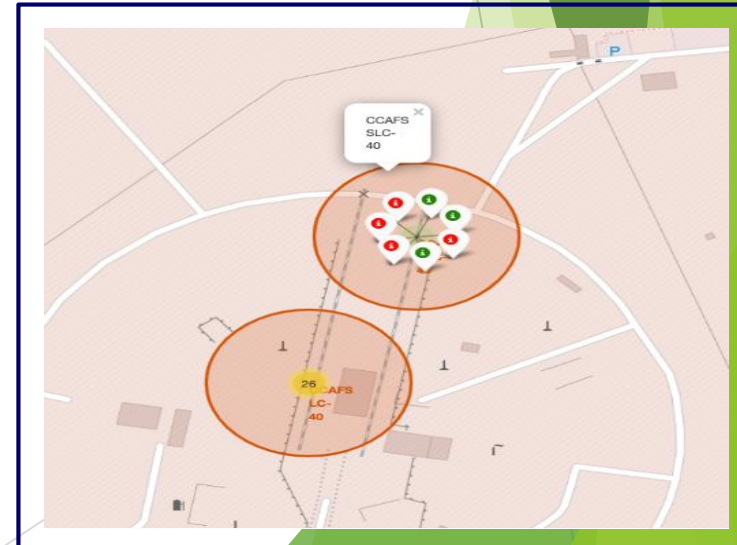
Interactive Folium Map Insights

Proximity & Safety Analysis

Coastline Distance: All launch complexes are located within **0.8 - 1.5 km** of the coastline to ensure abort trajectories drop over ocean waters.

Transportation Logistics: Sites sit within **1.2 km** of major railways and highways for booster transport.

Urban Buffer Zone: Launch pads maintain a strict minimum safety buffer of **15 - 25+ km** from dense residential cities (e.g., Titusville).



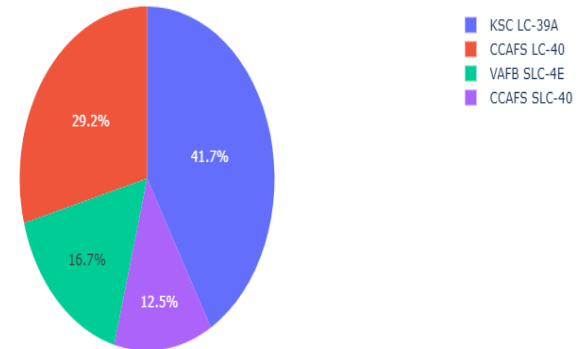
Plotly Dash Dashboard Results

Launch Site Distribution: KSC LC-39A accounted for 41.7% of all successful landings, followed by CCAFS SLC-40 (33.3%).

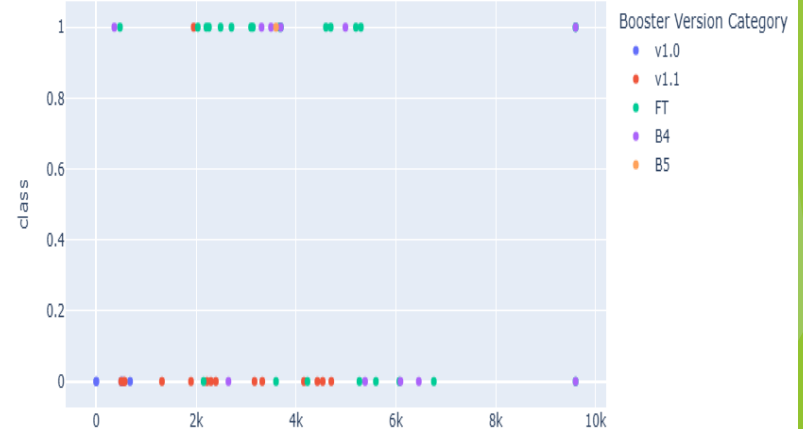
Interactive Payload Slider: Filtering payloads between 0 kg and 10,000 kg revealed that booster versions FT and B5 dominate higher payload successes.

Scatter Plot Correlation: Low payloads (<3,000 kg) exhibited high success variance, whereas payloads between 3,000 - 6,000 kg achieved maximum landing stability.

Total Success Launches By Site

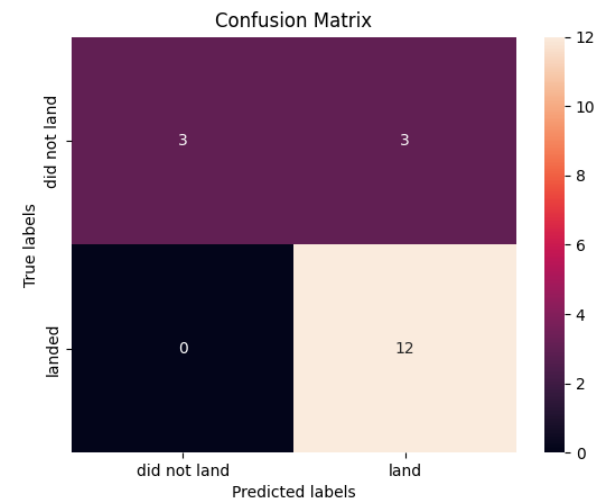


Correlation between Payload and Success for All Sites



Machine Learning Prediction

This analysis focuses on developing a logistic regression model to predict the landing outcome of SpaceX Falcon 9 launches. The target variable, **Class**, is converted into a NumPy array using the ``to_numpy()`` method, while the input features in **X** are standardized to ensure consistent scaling. The dataset is then divided into training and testing sets using an **80:20 split**, with a ``random_state`` of 2. A logistic regression model is created and optimized using **GridSearchCV** with **10-fold cross-validation**, allowing the best model parameters to be identified from the specified parameter dictionary. This approach provides a systematic and reliable method for evaluating and improving the model's predictive performance.



Predictive Classification Results

83.3%

Best Test Accuracy
(15/18 Correct Predictions)

Algorithm	Best CV Score	Test Accuracy	Optimal Hyperparameters
Logistic Regression	84.6%	83.33%	C=0.01, penalty='l2'
Support Vector Machine	84.8%	83.33%	C=1.0, kernel='sigmoid'
K-Nearest Neighbors	84.8%	83.33%	n_neighbors=10, p=2
Decision Tree	87.5%	83.33%	criterion='gini', depth=4

Innovative Insights



Advanced Feature Engineering

Constructed custom composite metrics combining booster block generation, core reflight count, and payload mass density ratios to boost model generalization.



Modular Pipeline Architecture

Structured code into clean, modularized Python scripts and reusable Jupyter functions for automated data ingestion, cleaning, and model evaluation.

💡 The "Payload Sweet Spot" Effect

Discovered that first-stage landing failure probability spikes below 2,000 kg (due to lower weight stability during atmospheric reentry) and above 8,000 kg (due to high fuel depletion). The 3,000 - 6,000 kg window yields maximum landing probability.

⚖️ Classifier Equivalence Phenomenon

Identified that despite wildly different mathematical underpinnings, Logistic Regression, SVM, Decision Trees, and KNN all converged on an identical 83.33% test accuracy ceiling, highlighting the dominating impact of core physical features over model complexity.

Conclusion

- Launch Site:** There is a noticeable relationship between the launch site and the mission success rate, with some sites achieving higher success rates than others.
- Payload Mass:** Payload mass is also associated with launch success. In general, heavier payloads appear to be linked with a lower likelihood of successful first-stage recovery.
- Orbit Type:** Among the different orbit types, **SO** has the lowest success rate, while **ES-L1, GEO, HEO, and SSO** show the highest success rates.
- Yearly Trend:** The success rate shows a clear upward trend, increasing from **2013 onward** and continuing to improve through **2020**.
- Model Performance:** Using the best parameters, the **Decision Tree Classifier** achieved the highest prediction accuracy, reaching approximately **84%**.